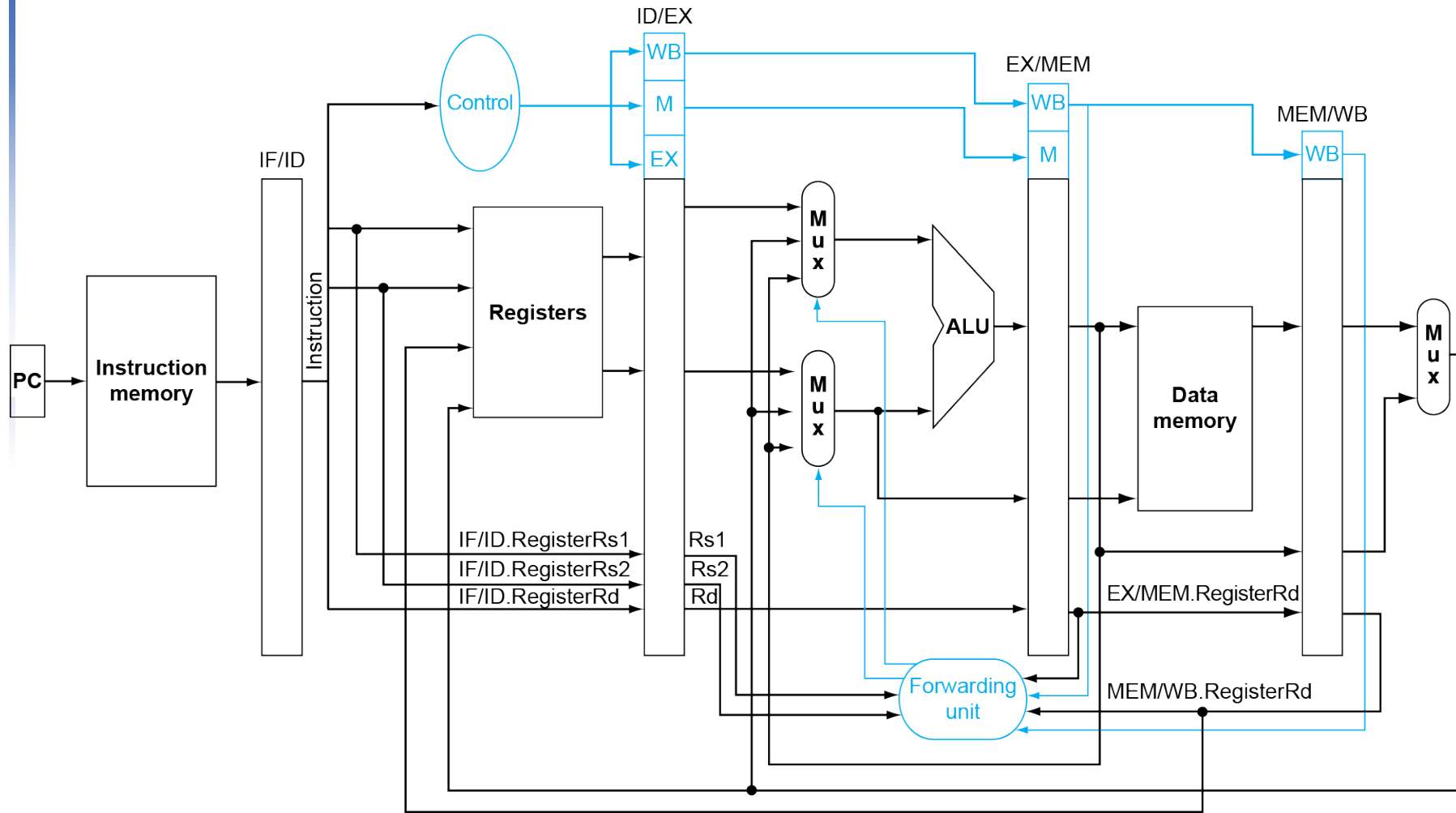


Chapter 4

The Processor

Revision: Forwarding



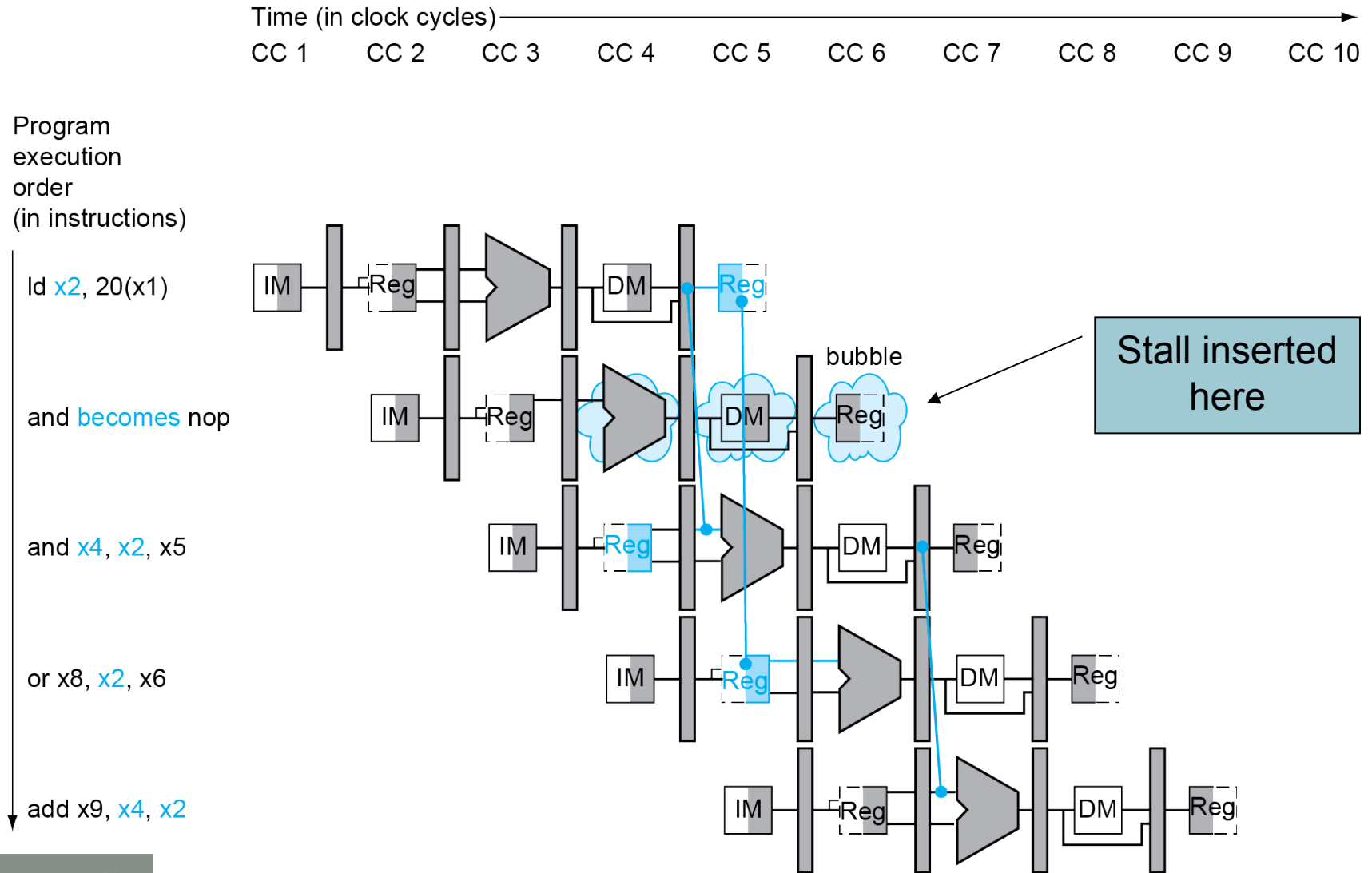
Load-Use Hazard Detection

- Check when using instruction is decoded in ID stage
- ALU operand register numbers in ID stage are given by
 - IF/ID.RegisterRs1, IF/ID.RegisterRs2
- Load-use hazard when
 - ID/EX.MemRead and
((ID/EX.RegisterRd = IF/ID.RegisterRs1) or
(ID/EX.RegisterRd = IF/ID.RegisterRs2))
- If detected, stall and insert bubble

How to Stall the Pipeline

- Force control values in ID/EX register to 0
 - EX, MEM and WB do nop (no-operation)
- Prevent update of PC and IF/ID register
 - Using instruction is decoded again
 - Following instruction is fetched again
 - 1-cycle stall allows MEM to read data for `ld`
 - Can subsequently forward to EX stage

Load-Use Data Hazard



MK



Stalls and Performance

The BIG Picture

- Stalls reduce performance
 - But are required to get correct results
- Compiler can arrange code to avoid hazards and stalls
 - Requires knowledge of the pipeline structure